Vortical Enclosure Dynamics, Vol. 1

Foundations

Independent Research — 2026

Yuhki Endoh

April 30, 2026

Contents

1 Introduction

Most physical theories begin from assumed structure: a spacetime manifold, a Lagrangian density, or a symmetry group. In all such cases, the framework presupposes what it seeks to explain.

VED takes a different approach. Starting from the single statement *there is difference*, the framework derives time, space, motion, and persistence as necessary consequences of structural asymmetry — without importing any external scaffolding.

The central object is the **causal log matrix** C_{ij}, defined on a discrete network of nodes. C_{ij} captures not the total history of causal influence between nodes, but the portion of that history that is selectively retained as persistent structure. From C_{ij}, a single self-referential equation generates the entire chain:

$$\text{difference} \to C_{ij} \to \rho \to \tau \to d_{ij} \to J \to \omega \to L \tag{1}$$

See Fig. 1 for a visual summary.

This framework is not presented as a complete predictive theory, but as a minimal structural derivation of physical organization. It is a structural proposal: a demonstration that the minimum ingredients required for physical structure are considerably smaller than those assumed in current foundational approaches. The aim is coherence, not closure.

Organization. Section 2 states the minimal axiom and traces its immediate consequences. Section 3 introduces the causal log and derives time. Section 4 derives spatial distance from the symmetric component of C_{ij}. Section 5 develops flow and vorticity from the antisymmetric component. Section 6 introduces closure as the minimal persistent structure. Section 7 establishes the phase structure of the network. Sections 8 and 9 discuss implications and open problems.

2 The Minimal Axiom

VED is founded on a single, irreducible statement:

There is difference.

This is not a metaphor or a philosophical preamble. It is a structural condition. A perfectly symmetric system produces no gradients, no flows, and no observable events. Without asymmetry, nothing happens. The overall structure is summarized in Fig. 1.

Therefore:

Proposition 2.1. *An observable world requires the existence of difference.*

This is not an empirical claim; it is a structural necessity. If the world were completely symmetric, every state would be indistinguishable from every other, and the concept of observation would be undefined.

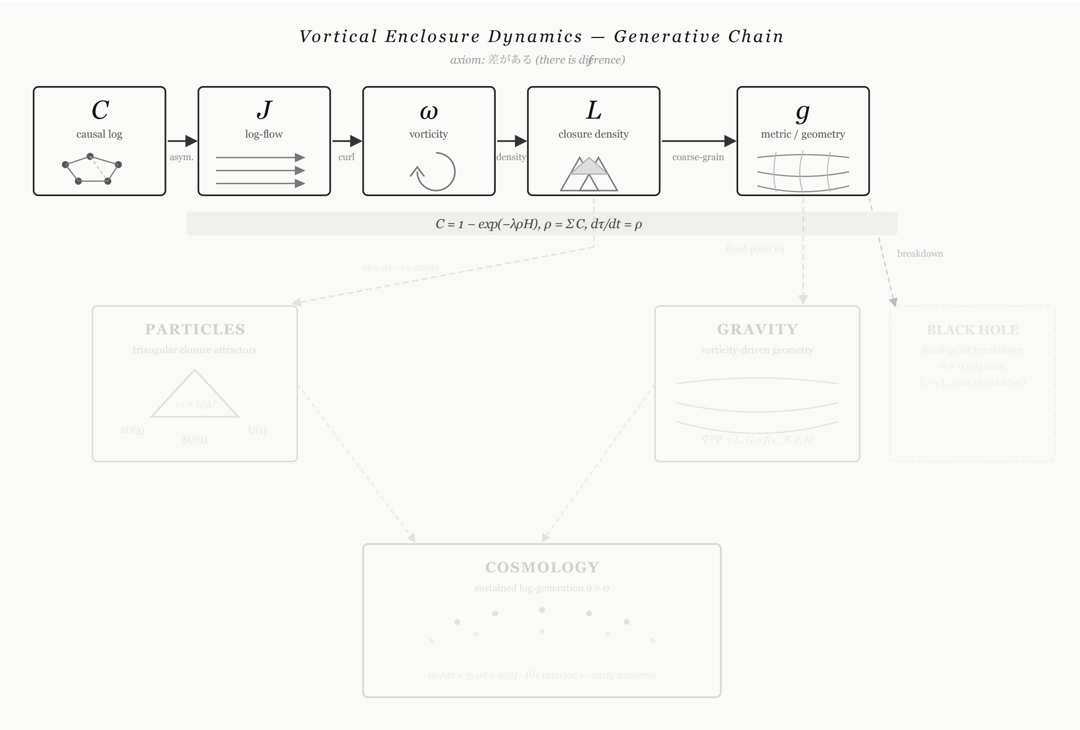

Figure 1: Generative chain of Vortical Enclosure Dynamics. Difference induces causal logs C_{ij}, which accumulate into density ρ, generating time τ. Connectivity defines distance d_{ij}, enabling flow J, which produces vorticity ω and leads to closure L.

2.1 From Difference to Dynamics

Given the existence of difference, a necessary chain follows:

$$\text{difference} \;\Rightarrow\; \nabla\delta \neq 0 \;\Rightarrow\; J \sim -\nabla\delta \;\Rightarrow\; \text{flow} \;\Rightarrow\; \text{vortex} \;\Rightarrow\; \text{closure}$$

Asymmetry implies a gradient; a gradient drives a flow; a sustained flow generates circulation; circulation stabilizes into closures. This chain is not imposed — it is the necessary consequence of the existence of asymmetry.

2.2 The Non-Closure Constraint

If a closure were complete — if all structural differences were eliminated — all gradients would vanish and dynamics would cease. This contradicts the minimal axiom. Therefore the complete-closure limit cannot be treated as an ordinary attainable state of the generated description:

$$\boxed{L < L_{\max} \quad \text{at all times.}} \tag{2}$$

In this volume we represent this non-closure constraint by the following effective closure barrier:

$$B(L) = -\kappa_B \log\left(1 - \frac{L}{L_{\max}}\right), \qquad L \to L_{\max} \;\Rightarrow\; B(L) \to \infty. \tag{3}$$

This term should be read as a phenomenological encoding of the fact that complete closure would erase the differences required for dynamics. It is not meant to be the final microscopic origin of non-closure. In Vol. 4, the same constraint is reinterpreted dynamically: the effective barrier may emerge from delayed rotational feedback, dissipation, and the instability of complete closure within a generated description.

Remark 2.2. The non-closure constraint has far-reaching consequences. In section 6, it provides an effective scaling for the mass spectrum of persistent structures. In Vol. 3, it is used to reinterpret the black hole center as an asymptotic closure limit rather than an ordinary singular object.

3 Causal Logs and the Generation of Time

3.1 History and Structure

We distinguish two directed relations between nodes i and j:

- $H_{ij} \geq 0$: the **causal history** — total accumulated influence from i to j, irreversible and unfiltered.
- $C_{ij} \in [0, 1)$: the **causal log** — the selected portion of that history that becomes persistent structure. C_{ij} represents the strength of retained influence: not what happened, but what continues to matter.

The fundamental distinction is:

$$H_{ij} \neq C_{ij}. \tag{4}$$

The world is not the sum of everything that occurred, but the structure of what was retained.

3.2 The Self-Referential Selection Equation

Selection is not externally imposed. It is generated by the structure itself. Define the **log density** at node i:

$$\rho_i = \sum_j C_{ij}. \tag{5}$$

The **causal log equation** is:

$$\boxed{C_{ij} = 1 - \exp(-\lambda\, \rho_i\, H_{ij}),} \tag{6}$$

where $\lambda > 0$ is the selection sensitivity, the sole external parameter.

Substituting equation (5) into equation (6) yields a self-referential fixed-point equation in ρ_i:

$$\rho_i = \sum_j \left[1 - \exp(-\lambda\, \rho_i\, H_{ij})\right].$$

The stable fixed points ρ_i^* of this equation correspond to persistent physical structure.

Remark 3.1. The self-referential structure is not circular. It is a fixed-point iteration: existing structure (ρ_i) governs how new history (H_{ij}) is selected into new structure (C_{ij}). Structure generates its own selectivity.

3.3 The Generation of Time

Time is not a background parameter. It is defined as the rate at which causal logs accumulate at a node:

$$\boxed{\frac{d\tau_i}{dt} = \rho_i. \tag{7}}$$

Several consequences follow immediately:

- Since $\rho_i \geq 0$ by construction, τ_i is monotonically non-decreasing. Irreversibility is built into the definition.
- Where $\rho_i = 0$, time does not advance.
- No separate thermodynamic argument for the arrow of time is required.

Time is the rate at which structure accumulates. Irreversibility is structural, not statistical.

3.4 The Log-Generation Rate

Taking the time derivative of ρ_i using equation (6) yields:

$$\dot{\rho}_i = \frac{\lambda\, \rho_i\, B_i}{1 - \lambda\, A_i}, \tag{8}$$

where $A_i = \sum_j e^{-\lambda\rho_i H_{ij}} H_{ij}$ and $B_i = \sum_j e^{-\lambda\rho_i H_{ij}} \dot{H}_{ij}$. The numerator $\lambda\rho_i B_i$ measures the rate of new history injection; the denominator $1 - \lambda A_i$ captures saturation by existing structure. This microscopic expression underlies the log-generation rate σ appearing in section 5.

4 Distance and the Emergence of Space

4.1 Decomposition of C_{ij}

The causal log matrix is generically asymmetric. We decompose it into symmetric and anti-symmetric parts:

$$S_{ij} = \tfrac{1}{2}(C_{ij} + C_{ji}), \qquad A_{ij} = \tfrac{1}{2}(C_{ij} - C_{ji}). \tag{9}$$

S_{ij} encodes mutual connectivity (the spatial component); A_{ij} encodes directed flow (the temporal and causal component). Time and space are not independently postulated; they arise as two aspects of the same object C_{ij}.

4.2 Spatial Distance

For a path $\gamma : i \to j$ through the network, define its **propagation strength**:

$$\tilde{S}_{ij} = \max_{\gamma:i\to j} \prod_{(ab)\in\gamma} S_{ab}.$$

The **distance** between nodes is:

$$d_{ij} = -\log \tilde{S}_{ij} = \min_{\gamma:i\to j} \sum_{(ab)\in\gamma} (-\log S_{ab}). \tag{10}$$

This is a shortest-path problem on a weighted network. The logarithm converts products to sums and ensures that distance accumulates additively along paths.

Proposition 4.1. *d_{ij} satisfies non-negativity and the triangle inequality, inherited from the path structure of the network.*

Distance is resistance to influence propagation. Space is not a background arena; it is a measure of how difficult information finds it to propagate between nodes.

4.3 Coarse-Graining and the Metric Tensor

In a coarse-grained limit, grouping nodes into spatial blocks and averaging produces a smooth effective distance:

$$d(x,y)^2 \approx g_{\mu\nu}(x)\,(x^\mu - y^\mu)(x^\nu - y^\nu).$$

$$g_{\mu\nu} = \text{effective description of } d_{ij} \text{ under coarse-graining.} \tag{11}$$

The metric tensor $g_{\mu\nu}$ is not a fundamental field. It is the coarse-grained approximation of the log-distance structure. No coordinate system is assumed; coordinates label coarse-grained blocks.

4.4 Time and Space Unified

The two emergent structures can be summarized as:

$$\frac{d\tau_i}{dt} = \rho_i \qquad \text{(time: quantity of logs)} \tag{12}$$

$$d_{ij} = -\log \tilde{S}_{ij} \qquad \text{(space: structure of logs)} \tag{13}$$

Time and space originate from different aspects of the same underlying object C_{ij}.

5 Flow and Vorticity

5.1 Log-Flow

The antisymmetric component of C_{ij} defines the **log-flow**:

$$J_{ij} = C_{ij} - C_{ji} = 2A_{ij}. \tag{14}$$

J_{ij} represents the net directional transfer of causal influence. It satisfies $J_{ij} = -J_{ji}$ by construction. Motion arises from asymmetry: where $J_{ij} = 0$, no net transfer occurs; where $J_{ij} \neq 0$, directed propagation occurs.

5.2 The Continuity Equation

In the coarse-grained limit, J_{ij} becomes a vector field $\mathbf{J}(x)$. The evolution of log density ρ satisfies:

$$\boxed{\partial_t \rho + \nabla \cdot \mathbf{J} = \sigma,} \tag{15}$$

where $\sigma \geq 0$ is the **log-generation rate**. σ is not an external source term. It is the rate at which the system fails to remain closed: wherever structure is incomplete, new logs are generated.

This is not a conservation law. The condition $\sigma = 0$ holds only in the classical fixed-point regime (section 7). Conservation is a property of a phase, not a fundamental principle.

$$\boxed{\sigma > 0 \iff \text{time is being generated.}} \tag{16}$$

5.3 Helmholtz Decomposition

The flow field admits a Helmholtz decomposition:

$$\mathbf{J} = \mathbf{J}_{\text{rot}} + \mathbf{J}_{\text{pot}},$$

where $\nabla \cdot \mathbf{J}_{\text{rot}} = 0$ and $\nabla \times \mathbf{J}_{\text{pot}} = 0$. The solenoidal component $\mathbf{J}_{\text{rot}}$ carries vorticity and is responsible for the formation of stable structure. The irrotational component $\mathbf{J}_{\text{pot}}$ carries the divergence associated with the log-generation rate σ.

5.4 Vorticity and Closure Density

Define the **vorticity** of the log-flow:

$$\omega = \nabla \times \mathbf{J}. \tag{17}$$

Vorticity measures local circulation. When $\omega = 0$, flows are irrotational and no stable structures form. When $\omega \neq 0$, circulation exists and flow lines can close on themselves.

The local intensity of vortical structure defines the **closure density**:

$$L \sim |\omega|. \tag{18}$$

L is the source of both particle structure (Vol. 2) and gravitational effects (Vol. 3).

5.5 The Generative Chain

The complete chain from asymmetry to structure is:

$$C_{ij} \xrightarrow{J=C-C^T} J \xrightarrow{\omega=\nabla\times J} \omega \xrightarrow{L\sim|\omega|} L.$$

6 Closure and Persistent Structure

6.1 What Is Closure

A **closure** is a configuration in which causal flow circulates along a loop, the loop reinforces itself through the self-referential selection equation equation (6), and the resulting pattern is dynamically sustained. A closure is not a boundary, but a circulation that sustains itself.

Without closure, flows dissipate. With closure, flow is trapped, gradients are maintained, and structure persists.

6.2 The Minimal Closure Is Triangular

The minimal loop that admits nontrivial closure must involve at least three nodes:

- **Two nodes**: the only cycle $i \to j \to i$ is purely oscillatory, with no internal degrees of freedom and no stable phase structure.
- **Three nodes**: the cycle $i \to j \to k \to i$ admits an oriented loop with internal area. The three edges span a two-dimensional phase space, allowing stable phase relationships.
- **Four or more**: any such cycle decomposes into triangular sub-loops. No new topological structure is introduced.

The minimal closure is triangular. This is a topological result, not an assumption.

6.3 Closure as a Dynamical Attractor

Define the **closure energy** of a configuration A:

$$E(A) = \alpha \left| \sum_{i\in A} \vec{e}_i \right|^2 - \beta\,\Phi(A) + \gamma|A| + B(L(A)), \tag{19}$$

where $\vec{e}_i$ are internal flux vectors derived from C_{ij}, $\Phi(A) = \sum_{i<j<k\in A} \phi_{ijk}$ is the non-degeneracy measure, and $B(L)$ is the effective non-closure barrier of equation (3).

Definition 6.1. *A **particle** is a local minimum of the closure energy $E(A)$.*

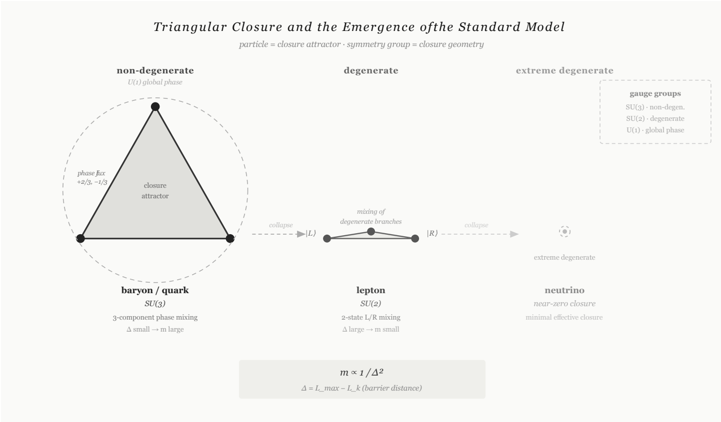

Figure 3: Triangular closure types and their correspondence to particle families. The three configurations illustrate non-degenerate, degenerate, and extreme degenerate closures as the simplest persistent enclosures, forming the basis for higher-order symmetries explored in subsequent work.

6.4 Closure Distance and Mass

The **closure distance** of a stable configuration:

$$\Delta_k = L_{\max} - L_k \tag{20}$$

measures how close the configuration lies to the complete-closure limit. The curvature of E at the minimum gives the effective mass. Within the phenomenological barrier approximation this yields:

$$\boxed{m_k \propto \left.\frac{d^2E}{dL^2}\right|_{L_k} \sim \frac{\kappa_B}{\Delta_k^2}.} \tag{21}$$

Here κ_B is an effective parameter of the projected closure-energy description. Vol. 4 gives a more dynamical interpretation in which the apparent barrier can arise from rotational feedback and dissipation rather than from a fundamental external potential.

> *Heavy structures sit close to the complete-closure limit. The world is made of incomplete closures.*

This minimal structure is illustrated in Fig. 3.

7 Phase Structure

The dynamics of the causal log network do not produce uniform behavior. Depending on the magnitude of the effective coupling, the network operates in qualitatively different regimes. Physical laws are properties of a specific phase, not universal principles valid across all parameter regimes.

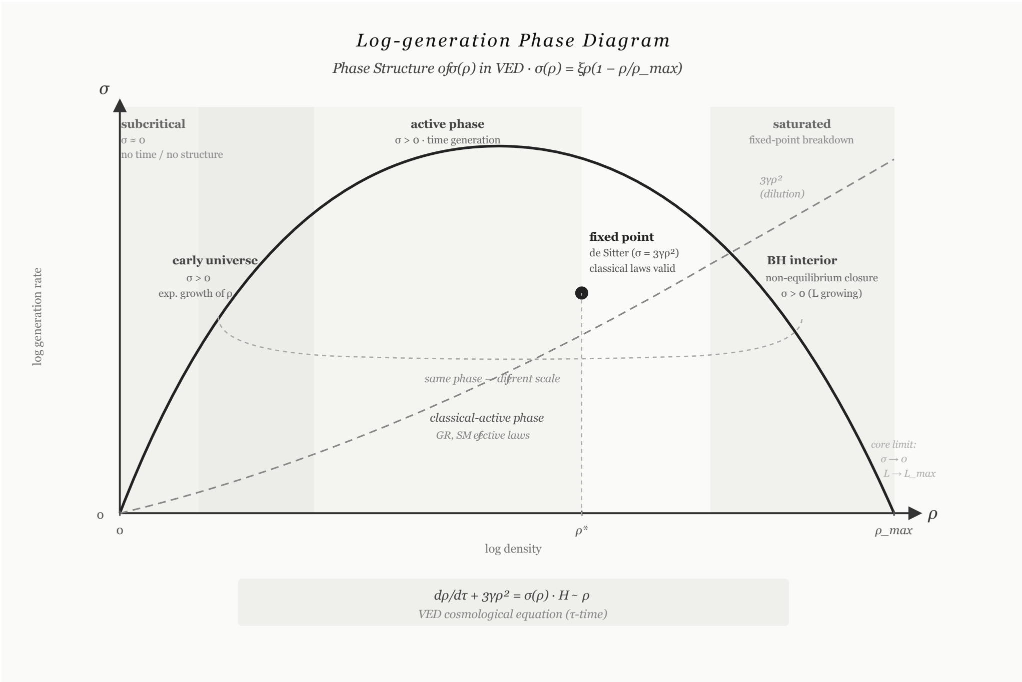

Figure 2: Phase structure as a function of density ρ. The system transitions across subcritical, critical, classical-active, and saturated regimes. Physical laws emerge as properties of stable phases.

7.1 The Control Parameter

The key parameter is:

$$\alpha_i = \lambda\, h_i, \qquad h_i = \sum_j H_{ij}, \tag{22}$$

where λ is the selection sensitivity and h_i is the total incoming causal history at node i. At $\alpha_i = 1$, a bifurcation occurs: below this value no log density can be self-sustained; above it, ρ_i rises and structure forms. The phase structure is shown in Fig. 2.

7.2 Four Phases

Phase	Condition	Character
Subcritical	$\alpha < 1$	$\rho \approx 0$; no time, no structure
Critical	$\alpha = 1$	Maximum susceptibility; bifurcation point
Classical-active	$\alpha \sim 1$	Stable closures; effective laws emerge
Saturated	$\alpha \gg 1$	$C_{ij} \to 1$; differences vanish

Table 1: Phase structure of the causal log network.

7.3 Emergence of Physical Laws

Within the classical-active phase, the following structures become stable:

- **Conservation laws**: $\sigma \approx 0$, so $\partial_t \rho + \nabla \cdot \mathbf{J} \approx 0$.

- **Gauge symmetries**: closure configurations stabilize with well-defined internal degrees of freedom (Vol. 2).
- **Effective geometry**: coarse-grained distance d_{ij} becomes approximately smooth, admitting a metric tensor $g_{\mu\nu}$ (Vol. 3).

Conservation, symmetry, and geometry are properties of the classical-active phase — not axioms. They emerge within it and break down outside it.

7.4 Extreme Regimes as Phase Transitions

The framework naturally accommodates extreme regimes:

- **Big Bang**: transition from subcritical to active phase. Structure bootstraps from a region where α crosses the critical threshold — not from a geometric singularity.
- **Black hole interiors**: transition from classical-active to non-equilibrium phase. The interior is not a singularity; it is a regime where fixed-point conditions break down and $\sigma > 0$ reignites (Vol. 3).

8 Discussion

VED proposes that the foundational primitives of physics — spacetime, fields, and symmetry groups — are not inputs but outputs. They emerge from the dynamics of a self-referential selection process operating on a discrete causal network. The present work should be understood as a structural proposal, not a finalized physical theory. Several aspects of this proposal merit explicit discussion.

Relation to existing frameworks. VED does not quantize general relativity, nor does it discretize quantum field theory. Its closest structural relatives are causal set theory [**?** **?**], in which spacetime is replaced by a partial order on discrete events, and relational quantum mechanics [**?**], which locates physical quantities in relations between systems rather than in absolute values. VED differs from both in that it derives spatial and temporal structure from a single self-referential equation, without postulating an underlying causal order.

Status of conservation laws. The continuity equation equation (15) does not impose conservation. Conservation ($\sigma = 0$) is a special case that holds in the classical-active phase. This is a departure from standard formulations, where conservation is an axiom or a consequence of symmetry (Noether's theorem). In VED, Noether's theorem applies within the classical phase as an effective statement; it does not hold in general.

The Lagrangian as an effective description. An early version of this framework attempted a Lagrangian formulation. This was abandoned for two reasons: (i) the Lagrangian principle $\delta S = 0$ presupposes a reversible time parameter t, but in VED, time τ must be derived, not assumed; (ii) the non-zero log-generation rate σ is structurally incompatible with a variational principle that requires conservation. The Lagrangian formalism is, however, recoverable as an effective description within the classical-active phase, where $\sigma \approx 0$.

Open problems. The framework presented here is structurally complete but quantitatively open. Key open problems include:

1. Derivation of the specific value of λ from first principles.
2. Proof of existence and uniqueness of fixed points for generic H_{ij}.
3. Quantitative derivation of mass ratios from Δ_k (Vol. 2).
4. Exact (non-linear) Schwarzschild solution from the fixed-point equation (Vol. 3).
5. Microscopic computation of Newton's constant G from λ, κ_B, and network density (Vol. 3).

9 Conclusion

We have presented a structural framework deriving time, space, and persistent physical structure from a single axiom: *there is difference*.

The central result is the generative chain equation (1), in which the causal log matrix C_{ij}, defined by the self-referential equation equation (6), produces: observational time τ (equation (7)), spatial distance d_{ij} (equation (10)), log-flow J (equation (14)), vorticity ω (equation (17)), closure density L (equation (18)), and the mass spectrum of persistent structures (equation (21)).

The system exhibits four phases (table 1). The observable universe is identified with the classical-active phase, in which conservation laws and effective geometry emerge as stable attractors. Physical laws are phase-dependent, not universal.

The world is not given. It is constructed. It persists because it never fully closes.

Volumes 2 and 3 develop the consequences of this framework for the Standard Model and for gravity and cosmology, respectively.

A Notation and Symbol Reference

The following table summarizes the principal symbols used in this volume.

Symbol	Definition	Introduced
H_{ij}	Causal history ($i \to j$ influence strength)	section 3
C_{ij}	Causal log (selected structure)	equation (6)
ρ_i	Log density $= \sum_j C_{ij}$	equation (5)
λ	Selection sensitivity (sole external parameter)	equation (6)
τ_i	Observational time ($d\tau_i/dt = \rho_i$)	equation (7)
t	Ordering parameter (not observational time)	section 3
S_{ij}	Symmetric component of C_{ij} (spatial)	equation (9)
A_{ij}	Antisymmetric component of C_{ij} (causal)	equation (9)
$\tilde{S}_{ij}$	Maximum-path propagation strength	section 4
d_{ij}	Spatial distance $= -\log \tilde{S}_{ij}$	equation (10)
$g_{\mu\nu}$	Effective metric tensor (emergent)	equation (11)
J_{ij}	Log-flow $= C_{ij} - C_{ji}$	equation (14)
σ	Log-generation rate	equation (15)
ω	Vorticity $= \nabla \times \mathbf{J}$	equation (17)
L	Closure density $\sim \lvert\omega\rvert$	equation (18)
$L_{\max}$	Complete-closure limit	equation (2)
$B(L)$	Effective non-closure barrier	equation (3)
κ_B	Effective barrier strength	equation (3)
$E(A)$	Closure energy of configuration A	equation (19)
Δ_k	Closure distance $= L_{\max} - L_k$	equation (20)
m_k	Effective mass scaling $\propto \kappa_B/\Delta_k^2$	equation (21)
α_i	Control parameter $= \lambda h_i$	equation (22)

Table 2: Symbol reference for Vol. 1.